

AI TOOLS FOR RESEARCH

Prof. Sonali Agarwal | IIIT Allahabad



Background

Course Syllabus



M.Tech. & Ph.D. • 3 Credits • 0-1-4-0

**From individual AI tools to a
verifiable research workflow**

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Why this course?

Research is changing at three levels

ACCESS

More literature and data
than a researcher can
read

ASSISTANCE

AI can search,
summarize,
code, analyze and write

ACCOUNTABILITY

Scientific claims still
require
human verification

The course teaches how to gain speed without losing scientific rigour.

Course identity at a glance

LEVEL

M.Tech. + Ph.D.

CREDITS

3

L-T-P-S

0-1-4-0

ORIENTATION

Practice-
intensive

The syllabus is designed around hands-on research workflows, not only theory.

What background is expected?

PROGRAMMING

Basic Python
preferred

RESEARCH

Methodology +
academic writing

DATA

Basic statistics +
data analysis

SCHOLARSHIP

Digital libraries +
citation practice

**No prior experience with Large Language Models is
assumed.**

The course objective in one sentence

Use modern AI effectively and responsibly across the complete research lifecycle.

Discovery

Ideation

Reading

Writing

Coding

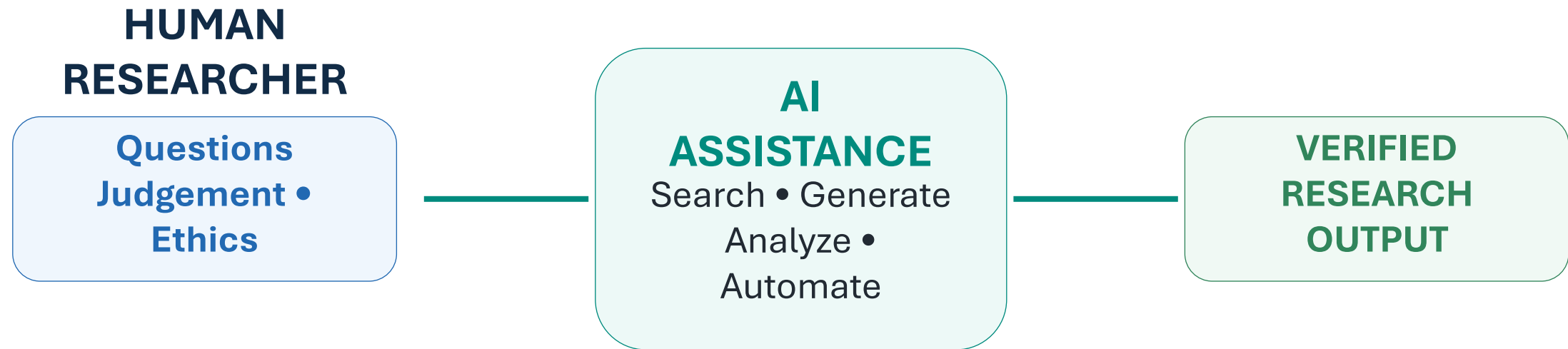
Analytics

Experimentation

Publication

...while developing critical judgement about hallucination, bias, citations, integrity and governance.

What does 'AI-augmented research' mean?



AI assists. Evidence and researcher judgement decide.

Course Outcomes: the learning journey



The outcomes move from understanding tools to integrating a complete ethical workflow.

CO1: Understand and evaluate

UNDERSTAND

Role of AI across the
research lifecycle

EVALUATE

Compare tools using
principled criteria

Key question: Which AI tool is appropriate for this research task, and why?

CO2: Conduct AI-assisted literature reviews

DISCOVER

Find candidate
literature

SCREEN

Select relevant
studies

SYNTHESIZE

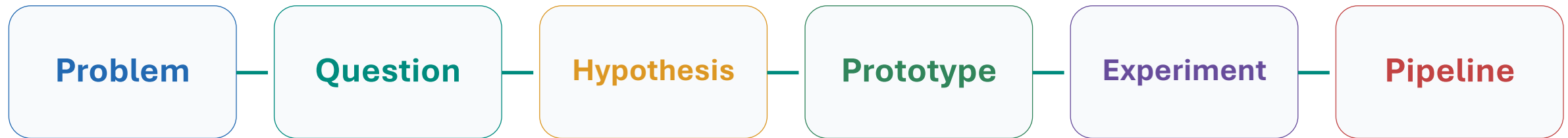
Compare and
organize
evidence

IDENTIFY

Research gaps

Systematic review ≠ asking an AI to 'summarize the literature'.

CO3: Formulate, code and experiment reproducibly



AI-assisted coding is useful only when the resulting workflow can be reviewed, tested and reproduced.

CO4: Design advanced AI research workflows

RAG

Retrieve evidence
before generation

AGENTS

Plan and execute
multi-stage tasks

MULTIMODAL

Work across text,
images, tables and
more

Goal: automate multi-stage tasks while preserving verification and reproducibility.

CO5: Critically assess and integrate

CONTENT

CODE

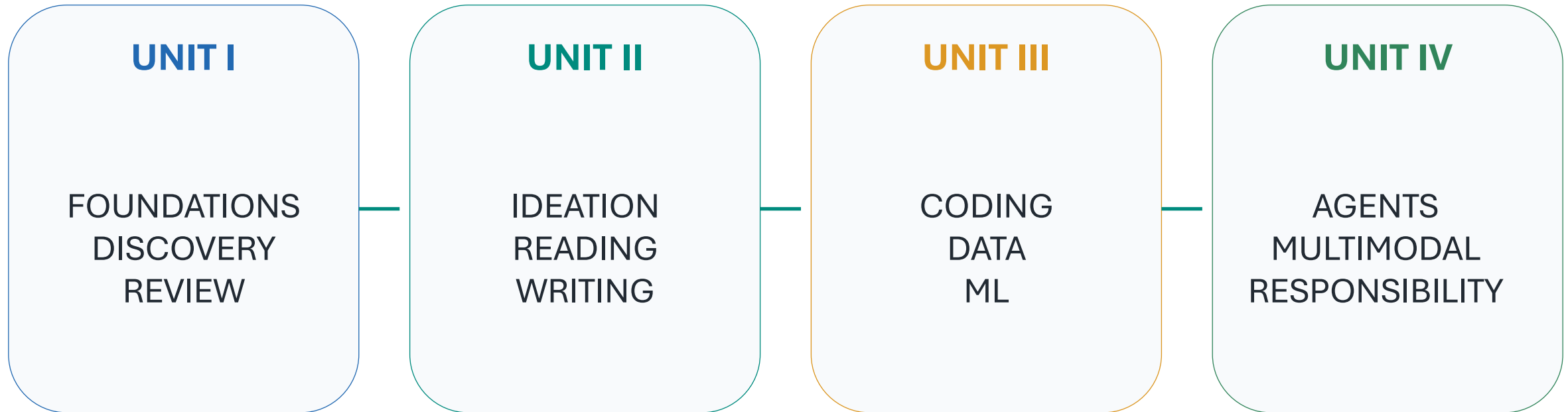
CLAIMS

ETHICS



**Complete, ethically grounded,
end-to-end AI-augmented research workflow**

Four units, one coherent story



Foundations → Research production → Computational workflows → Integration

Unit I begins with the foundations

RESEARCH EVOLUTION

Traditional → digital →
AI-assisted research

RESEARCH LIFECYCLE

Where AI can assist
and where humans
decide

AI LITERACY

LLMs • prompts •
context
reliability • limitations

Foundation before tools: understand what AI is doing to the research process.

Unit I: Prompt engineering + tool evaluation

A good research prompt is only half the problem.

SOURCE BASE

What information
can the tool access?

CAPABILITY

What can the
model actually do?

RELIABILITY

How consistently
can we trust it?

Tool selection should be principled, not based on popularity.

Unit I: Research discovery & knowledge exploration

**Scholarly
search**

**Citation
networks**

**Research
trends**

**Knowledge
maps**

**Semantic
Scholar**

**Research
Rabbit**

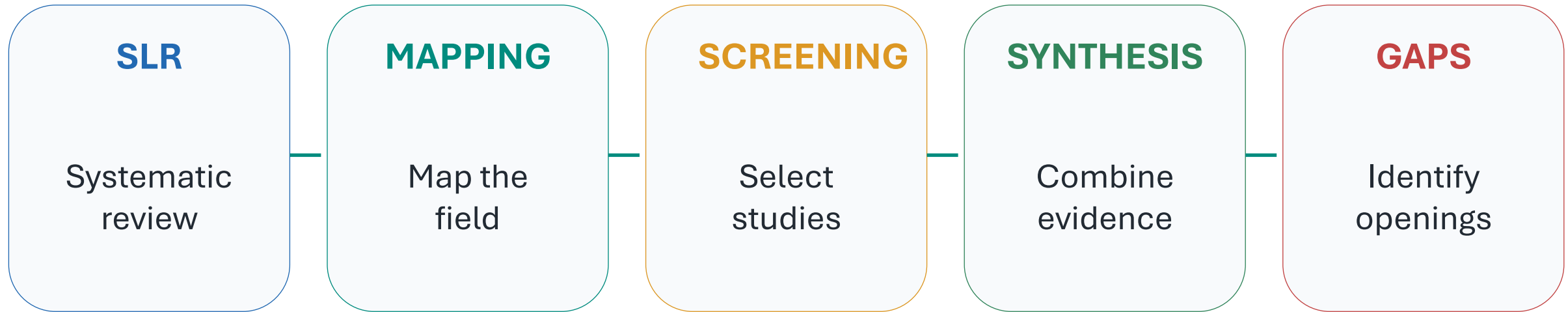
**Connected
Papers**

OpenAlex

Litmaps

Aim: move from a keyword list to an understanding of the research landscape.

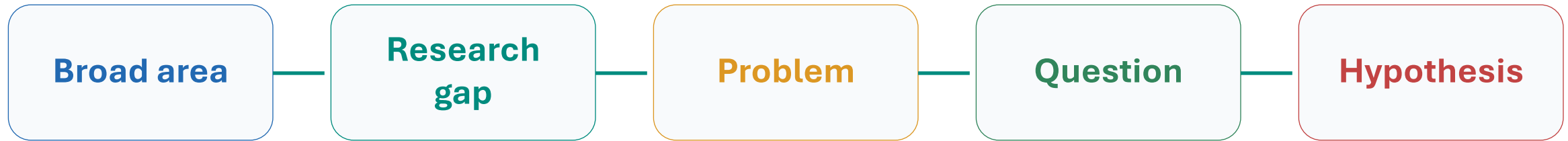
Unit I: AI-assisted literature review



Tools: Elicit • Consensus • SciSpace • Perplexity • etc.

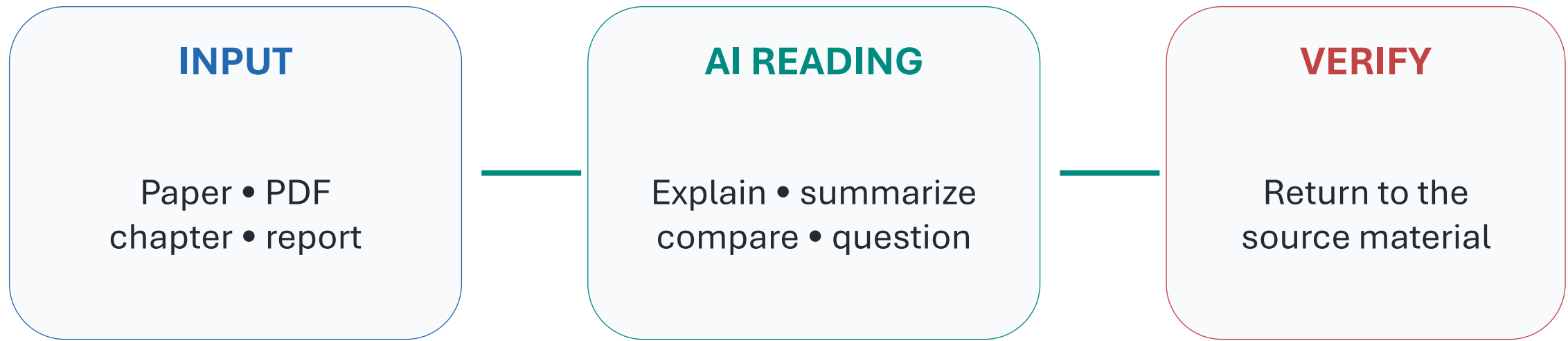
Critical check: hallucination and citation fabrication

Unit II: From an idea to a research question



AI can generate possibilities. The researcher must decide what is meaningful and testable.

Unit II: AI for academic reading



Tools: NotebookLM • Humata • ChatPDF • SciSpace

Unit II: AI for scientific writing

ABSTRACT

LITERATURE
REVIEW

METHODOLO
GY

RESULTS
INTERPRETATI
ON

CONCLUSION

Voice • originality • plagiarism avoidance • responsible revision

Tools: Jenni AI • Paperpal • Grammarly • general-purpose LLMs

AI-generated prose still requires author verification and responsibility.

Unit II: AI in scientific publishing

VENUE

Journal /
conference
selection

RESPONSE

Reviewer-
response
generation

LETTER

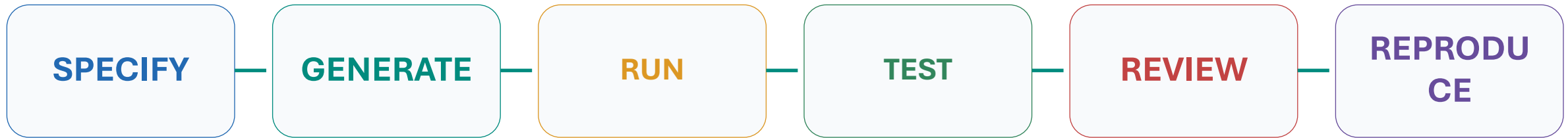
Cover-letter
preparation

SUBMISSION

Conference
support

AI can assist communication. It cannot decide whether the science meets a venue's standards.

Unit III: AI-assisted coding for research



Tools: GitHub Copilot • Cursor • Windsurf • Claude Code

Research code must survive testing, security checks and reproducibility.

What is 'vibe coding' in this syllabus?

RAPID

Conversational
prototyping

ITERATIVE

Specification-driven
refinement

TOOL-AUGMENTED

AI helps create and
modify software

Useful for prototypes. Dangerous when generated code is accepted without review.

Critical practices for vibe coding

REVIEW

TEST

VERIFY

DEPENDENCIES

TECHNICAL DEBT

REPRODUCIBILITY

Watch for silent errors • security flaws • unverified dependencies

Unit III: AI for data analytics & visualization



Tools: Julius AI • ChatGPT Advanced Data Analysis • Tableau AI • Power BI Copilot

The model may produce a chart. The researcher must judge whether the analysis is valid.

Unit III: AI for machine-learning research

AUTOML

Model selection +
training support

EXPERIMENTS

Compare runs and
configurations

TRACKING

Record metrics,
artifacts and models

Tools: AutoGluon • H2O AutoML • MLflow

Unit IV: Agentic AI for research automation



LangChain • LangGraph • CrewAI • AutoGen

Also: multi-agent systems • research assistants • workflow automation

More autonomy → more systematic verification

Unit IV: Multimodal AI + interoperability

TEXT

PDF

TABLES

IMAGES

AUDIO

VIDEO

CODE

Multimodal models: GPT-4o • Gemini • Claude • open VLMs

MCP Model Context Protocol

A2A Agent-to-Agent

Unit IV: Reproducible and responsible research

REPRODUCE

GitHub • Zenodo
MLflow • W&B • DVC

AUDIT

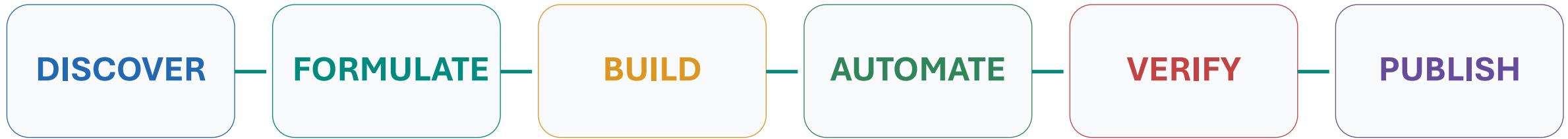
Hallucination • bias
citation fabrication

PROTECT

Copyright • plagiarism
integrity • governance

Responsible AI is embedded throughout the course, not added after the research is finished.

Where the course is heading



AI co-scientist systems • autonomous laboratories • end-to-end research workflows

**Final mindset:
Use AI to increase research capability, not to outsource scientific judgement.**

Learning resources in the syllabus

AI FOUNDATION

Russell & Norvig
Artificial Intelligence: A Modern
Approach

RESEARCH

Booth, Colomb & Williams
The Craft of Research

ML

Kelleher et al.
Predictive Data Analytics

LLMs

Tunstall, von Werra & Wolf
NLP with Transformers

Plus recent 2023–2026 surveys, official documentation and scholarly research resources.

AI TOOLS FOR RESEARCH

Week 1: Foundations of AI-Assisted Research

From Traditional Research to Human–AI Collaboration

“If AI can search, summarize, analyze data, write code and draft papers, what is the role of the researcher?”



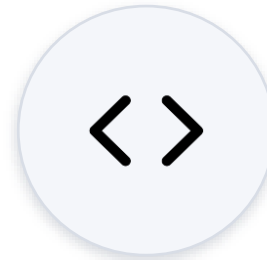
Search



Read



Analyze



Code



Write



Review

What Can AI Already Do for a Researcher?



**Discover
papers**



**Summarize
literature**



**Explain
difficult
concepts**



**Generate
search terms**



**Compare
studies**



**Extract
information**



**Generate /
debug code**



**Analyze
datasets**



**Suggest
visualizations**



**Assist
scientific
writing**



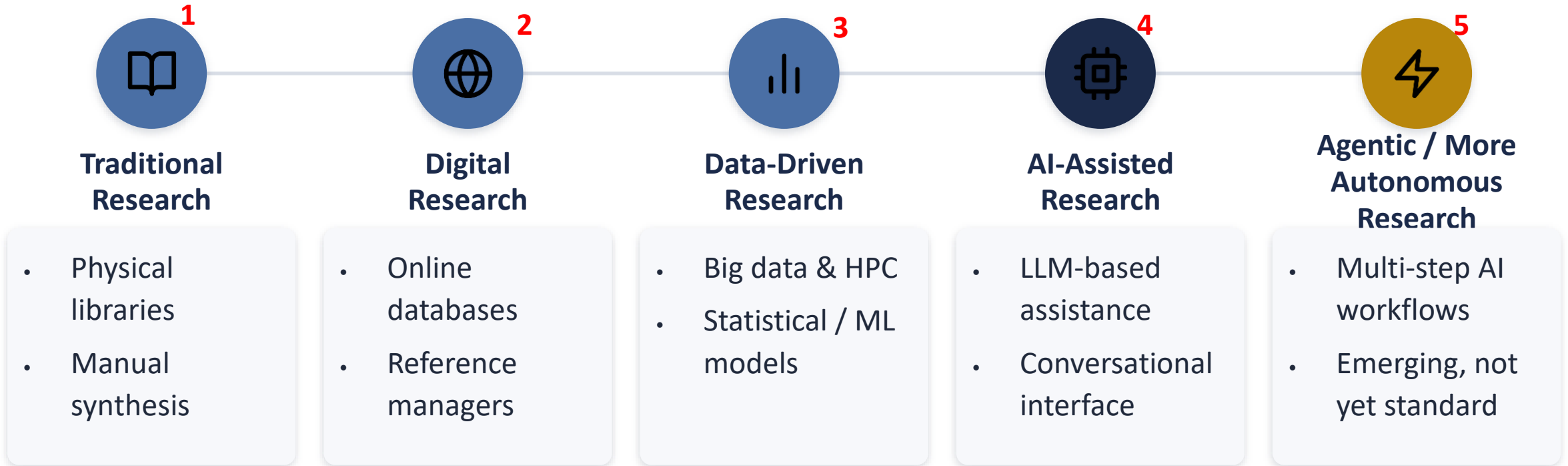
**Translate /
rewrite**



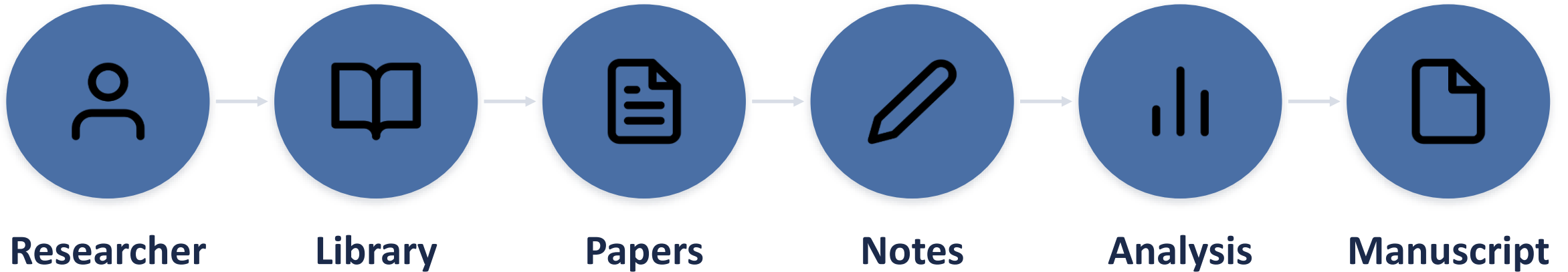
**Support peer-
review prep**

Capability $\neq$ Reliability

How Scientific Research Has Evolved



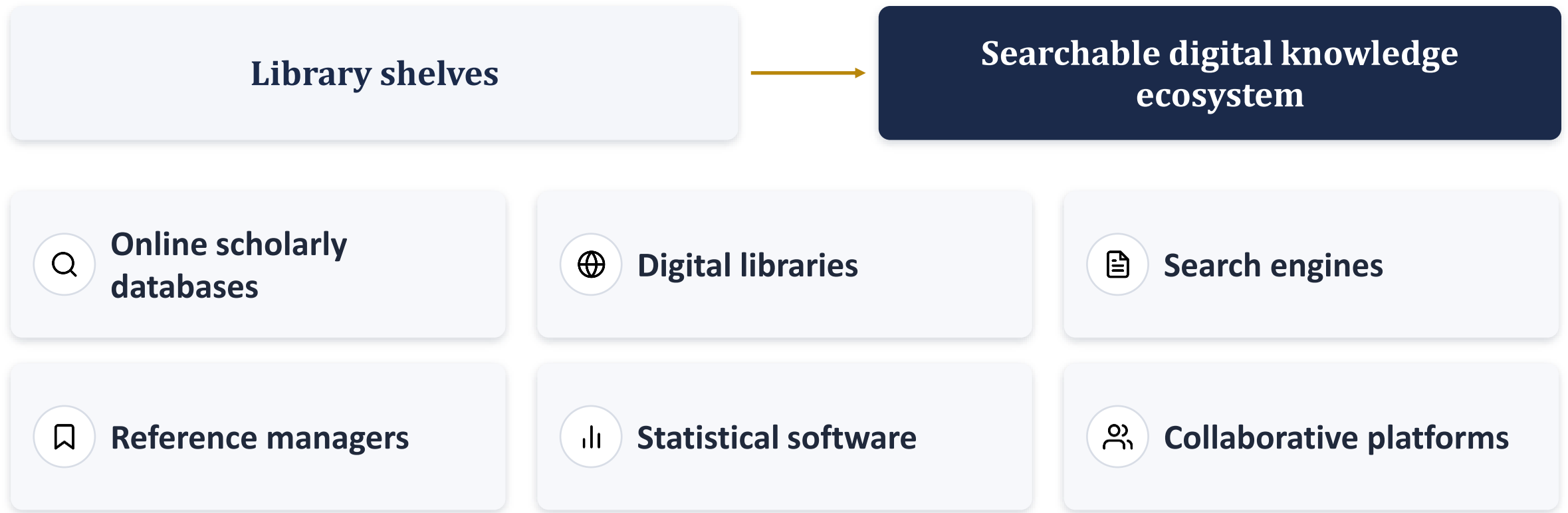
Traditional Research



- Physical literature search through library shelves and card catalogues
- Manual note-taking and hand-written or typed records
- A smaller, more locally accessible body of literature
- Manual or basic mechanical calculation
- Researcher-intensive synthesis - the scholar personally connects every idea

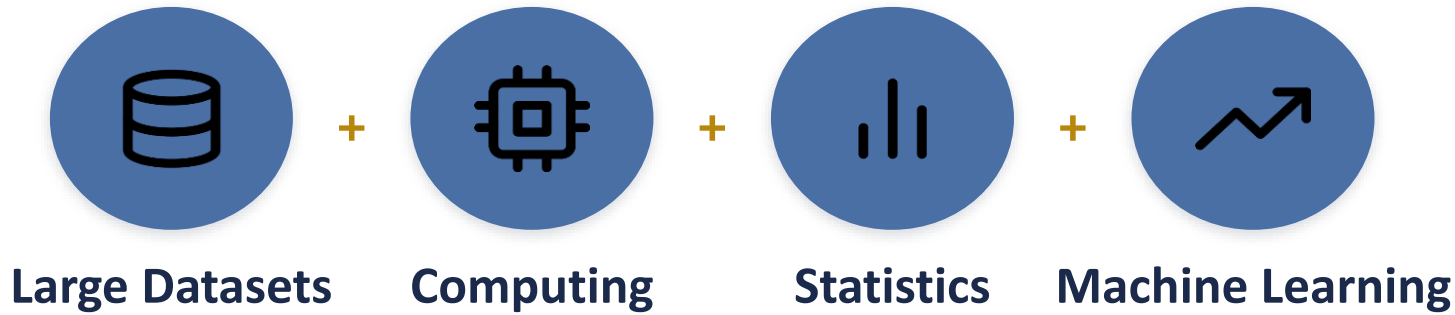
SECTION I · EVOLUTION OF SCIENTIFIC RESEARCH

Digital Research



Digitization primarily improved access, storage and computation - not the underlying logic of inquiry.

Data-Driven Research



Big data

Datasets far larger than any individual can review manually

Simulation

Modeling complex systems computationally before/instead of physical trials

Machine learning

Algorithms that detect patterns across large feature spaces

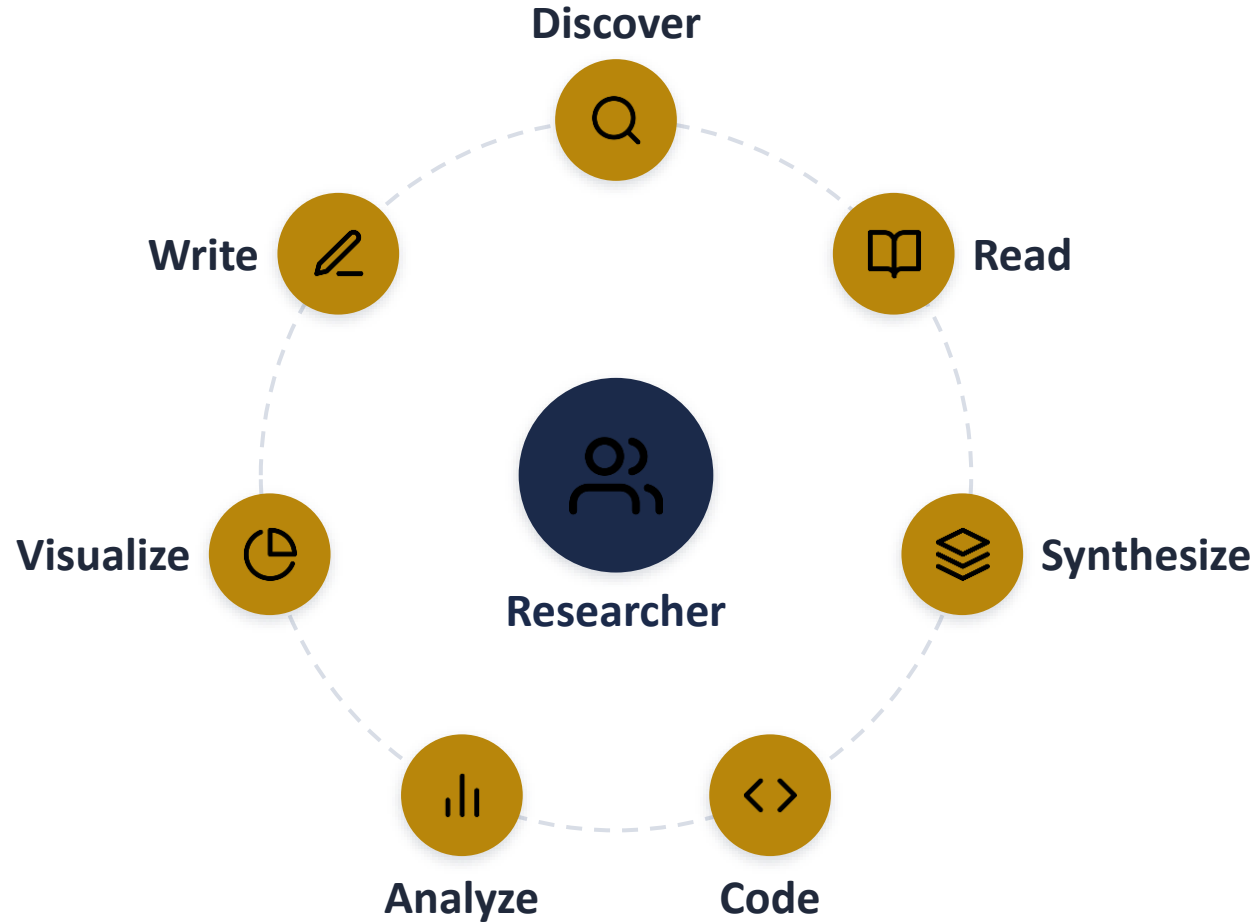
Cloud / HPC

On-demand computing power for large-scale processing

Automated experiments

Instrumentation that runs and records trials with less manual intervention

AI-Assisted Research



AI is increasingly becoming a general-purpose interface to research activities.

AI assistance does not transfer scientific responsibility from the researcher to the AI.

What Is Changing Fundamentally?

Traditional / Digital Workflow	AI-Assisted Workflow
Search keywords	Conversational / semantic discovery
Read papers individually	AI-assisted screening
Manual extraction	Structured extraction
Manual comparison	AI-assisted synthesis
Write code from scratch	AI-assisted coding
Static search	Interactive exploration

The bottleneck is shifting from “finding information” toward “evaluating information.”

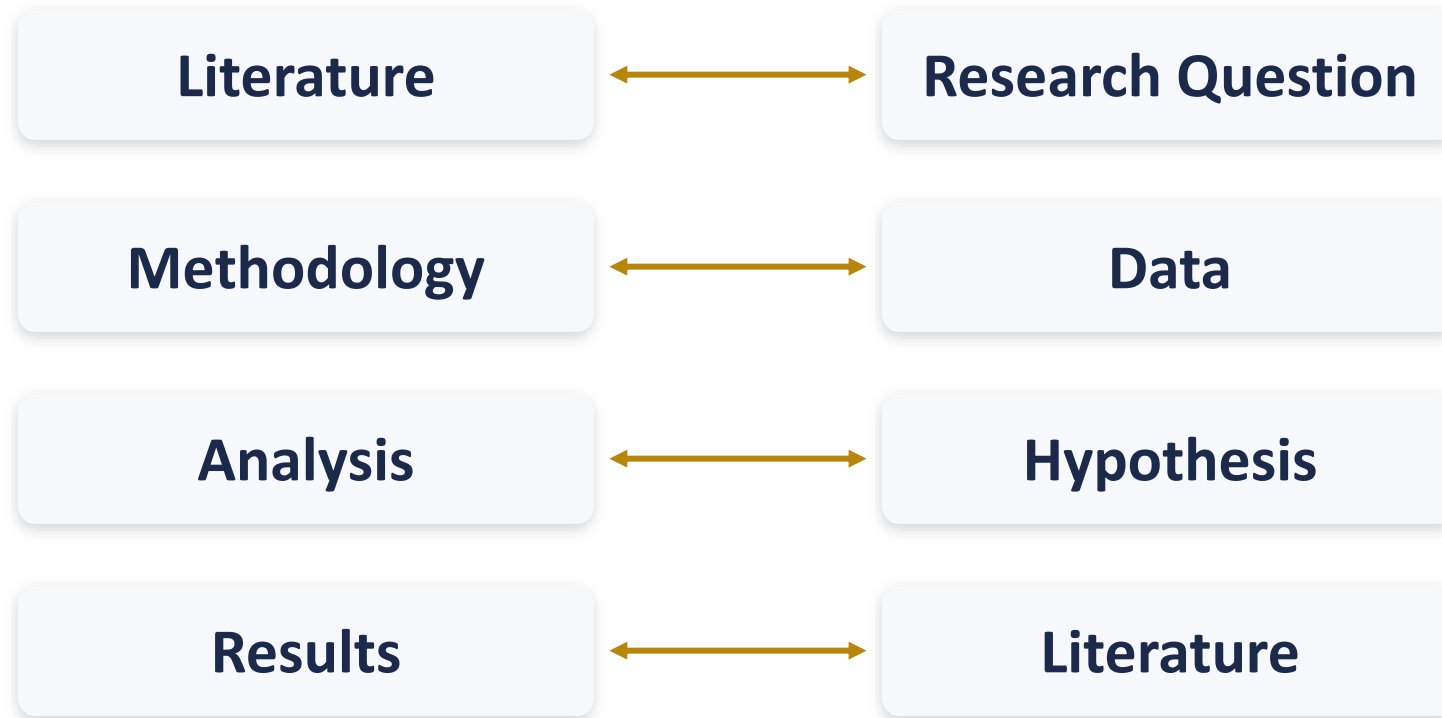
The Complete Research Lifecycle



The lifecycle is iterative, not strictly linear.

- Researchers move backward as often as forward
- A single project may revisit "literature" or "design" many times
- Publication is rarely the final step - it often raises new questions

Research Is Not a Straight Line



Real research involves repeated questioning, revision and verification.

Where Can AI Enter the Research Lifecycle?



AI can potentially support nearly every stage of the lifecycle.

“Does the ability to assist imply the authority to decide?”

No.

AI Support Across Research Stages

Research Stage	Possible AI Assistance	Human Verification Required
Problem discovery	Brainstorm directions	✓
Literature search	Find candidate literature	✓
Review	Summarize and compare	✓
Gap identification	Organize limitations	✓
Methodology	Suggest alternatives	✓
Coding	Generate / debug code	✓
Analysis	Assist statistical / computational work	✓
Interpretation	Generate possible explanations	✓
Writing	Draft / revise	✓
Publication	Language & reviewer-response support	✓

What Is AI Good At?



Speed



Scale



Repetitive processing



Summarization



Pattern identification



Generating alternatives



Formatting



Coding assistance



Language transformation

Labelled as potential strengths - not guarantees.

What Are Humans Still Responsible For?

✓ Selecting meaningful research problems

✓ Assessing scientific validity

✓ Identifying causal vs. correlational claims

✓ Interpreting significance

✓ Making final claims

✓ Understanding context

✓ Designing defensible methodology

✓ Ethical judgement

✓ Checking evidence

✓ Taking responsibility

The researcher remains epistemically responsible for the research claim.

Human vs AI: Complementary Strengths



AI

- Speed
- Scale
- Generation
- Search assistance
- Pattern processing
- Repetitive tasks



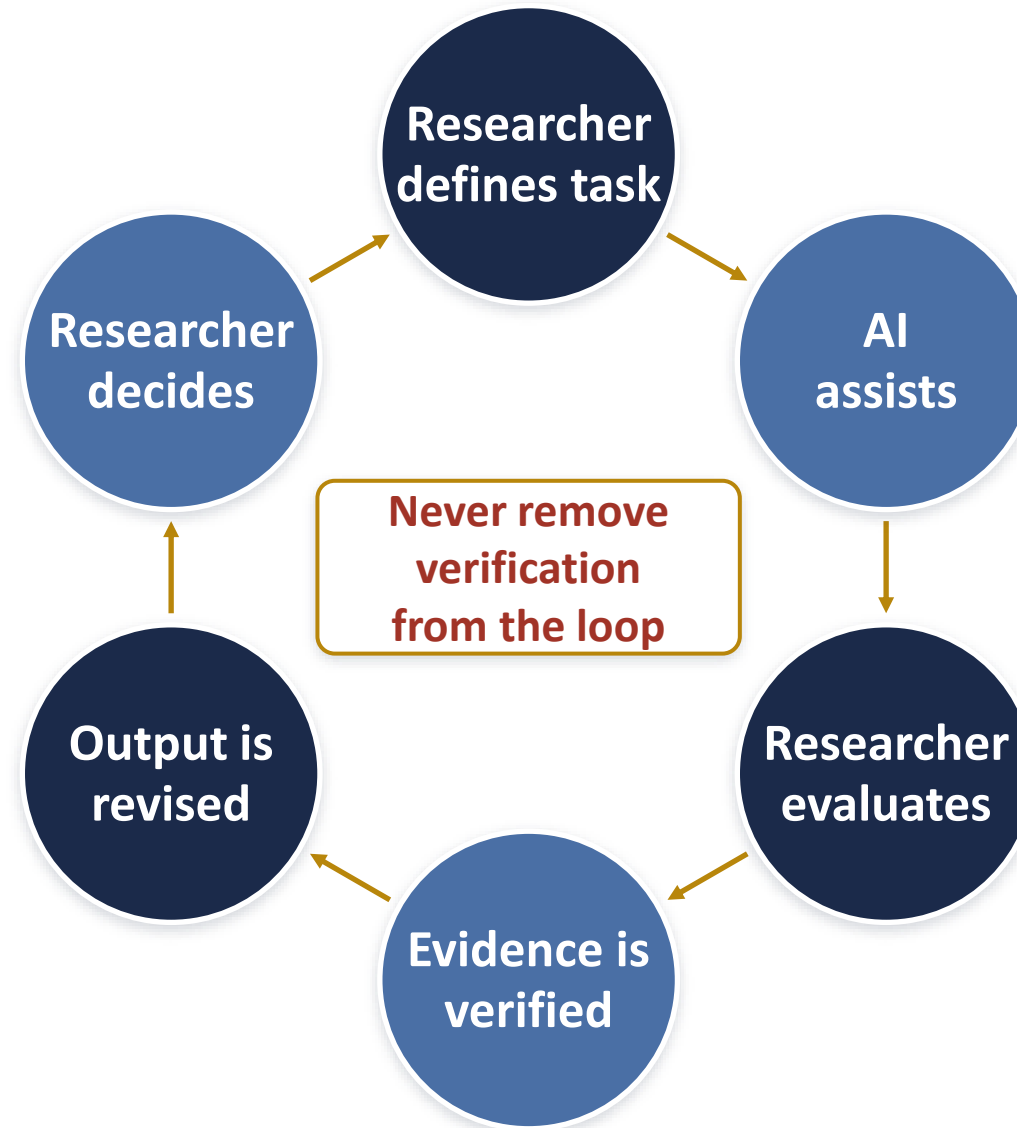
Human

- Context
- Judgement
- Scientific reasoning
- Ethics
- Meaning
- Accountability

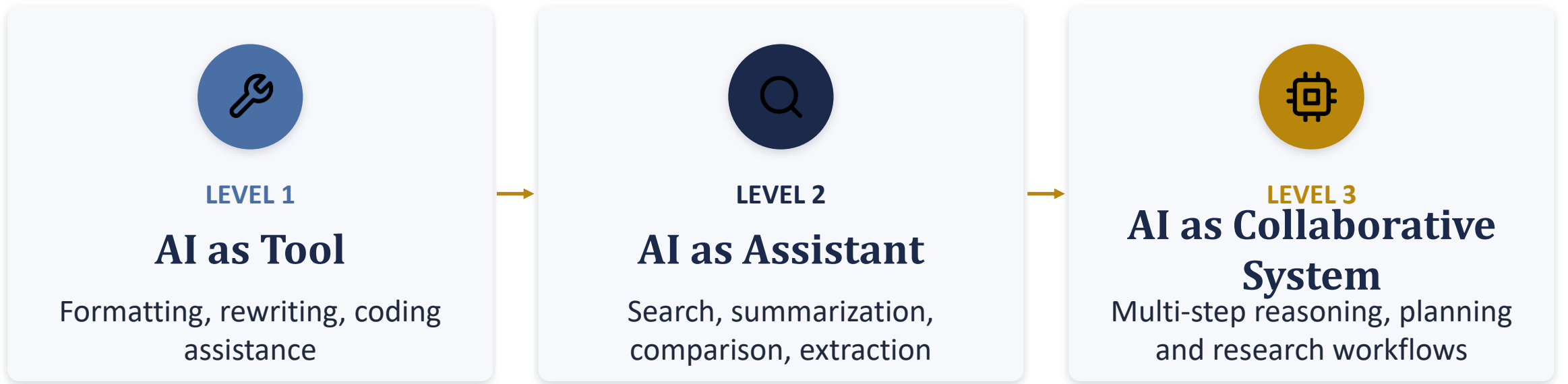
**Human
+
AI**

***Augmented
Research***

Human-in-the-Loop Research



Three Levels of AI Use

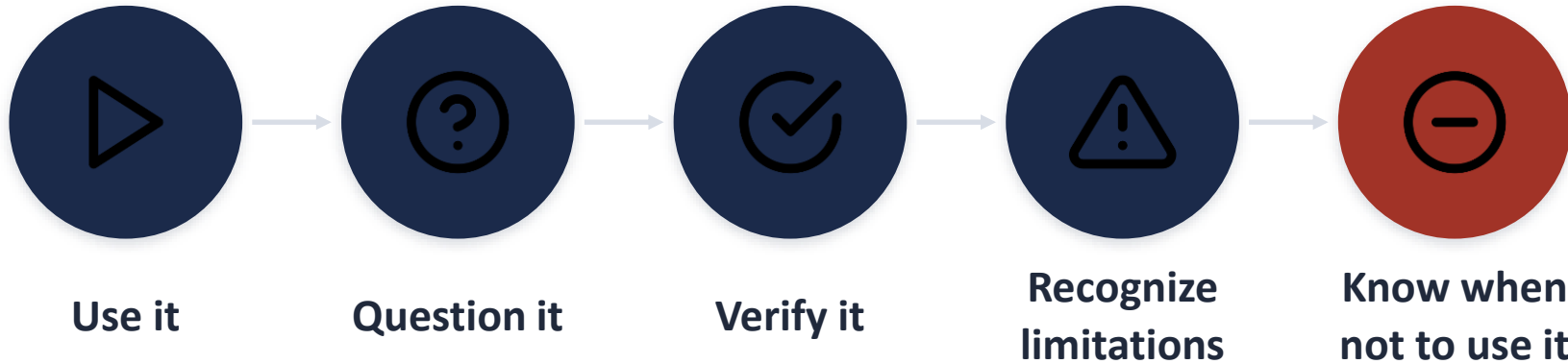


As autonomy increases, what should happen to verification?

Verification should become more systematic, not less.

What Does “AI Literacy” Mean for a Researcher?

AI literacy means understanding enough about AI to:



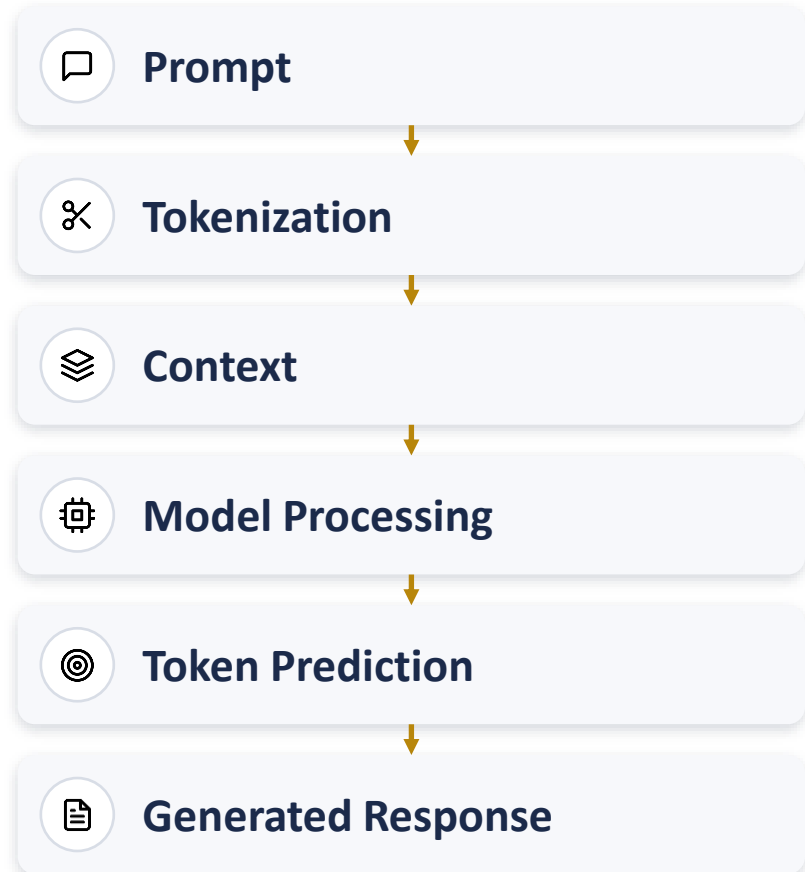
What Is a Large Language Model?



An LLM primarily generates responses based on learned patterns and the context it is given - it is not a database performing lookups.

Fluent language does not guarantee factual correctness.

How an LLM Interaction Works



Prompt

The instruction or question a researcher provides to the model.

Token

A small unit of text (often a word-piece) the model reads and generates.

Context

The full set of text currently visible to the model during this interaction.

Output

The generated response, built one predicted token at a time.

Context Window

A researcher's working desk

-  Your current prompt
-  Pasted documents / data
-  Earlier turns in this conversation

Context window = the information currently available to the model during this interaction.

Training Knowledge

What the model learned during training - broad, but frozen at a point in time.

Current Context

What you have placed in front of it this session - your prompt, files, chat history.

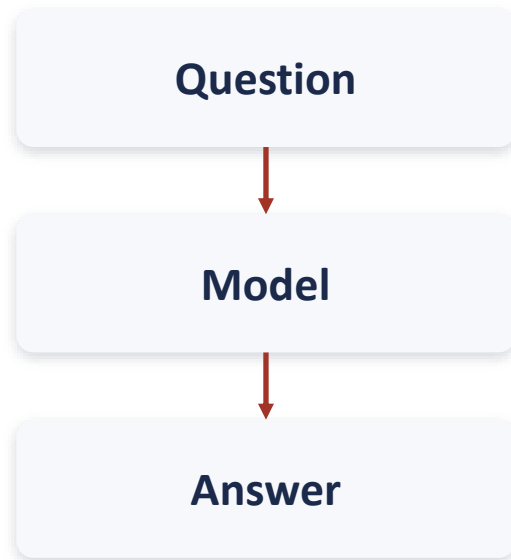
Retrieved Evidence

Specific documents fetched for this query, e.g. via search or RAG.

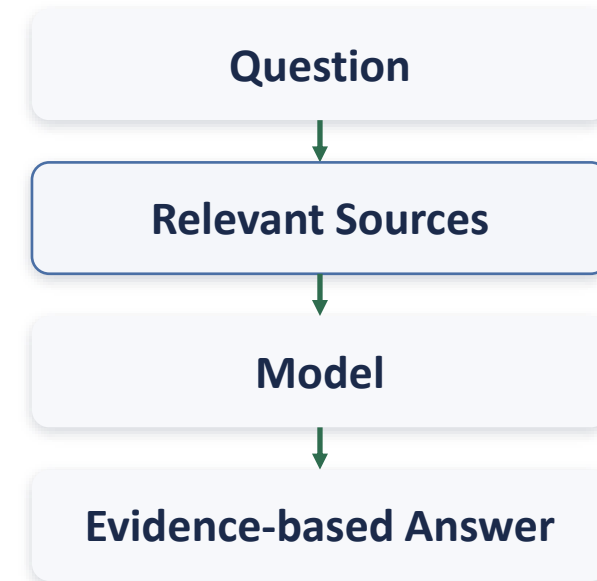
Grounding

“Where did this answer come from?”

UNGROUNDED RESPONSE

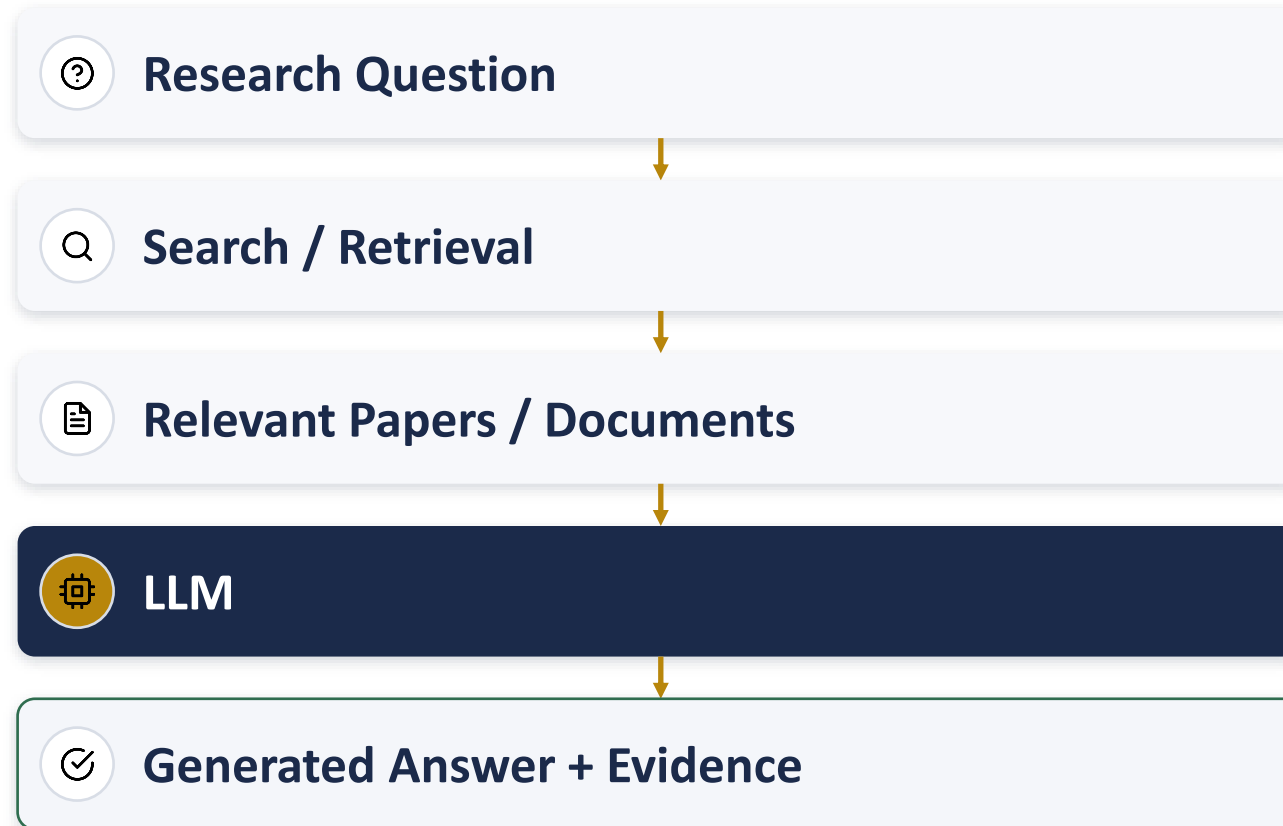


GROUNDING RESPONSE



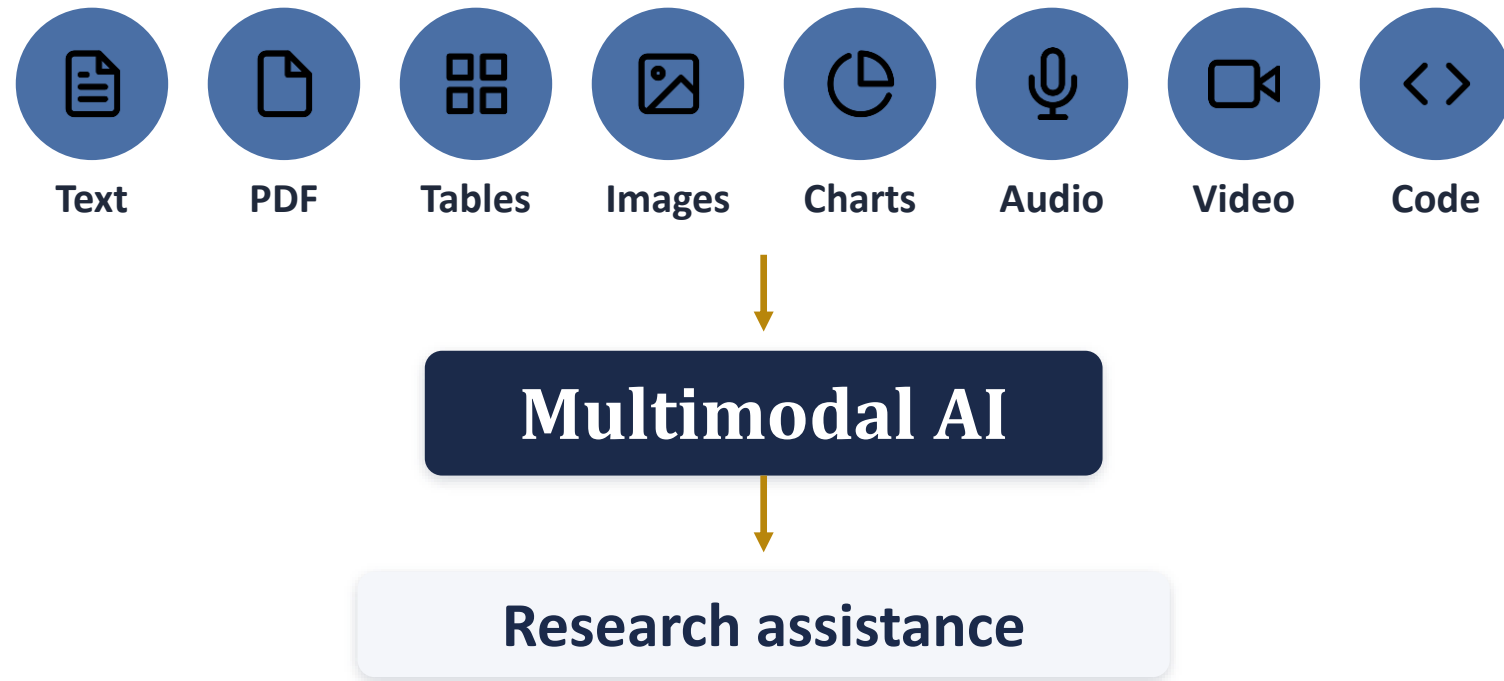
Grounding lets a claim be traced back to a specific source - essential for research use.

Retrieval-Augmented Generation (RAG)



RAG does not automatically make an answer correct - it makes evidence available for verification.

Multimodal AI for Research



- Interpreting scientific figures
- Analyzing images
- Reading PDFs
- Discussing datasets
- Explaining tables
- Generating code from descriptions

What Is an AI Hallucination?

A hallucination occurs when an AI generates information that appears plausible but is unsupported, inaccurate or fabricated.



Wrong Fact

A confidently stated detail that is simply incorrect.



Invented Citation

A reference to a paper, author or journal that does not exist.



Incorrect Interpretation

A real paper described in a way it does not actually say.

The Most Dangerous Research Hallucination

Citation Fabrication

Author	→	Title	→	Journal	→	Year	→	DOI
<i>Kapoor, R. et al. (2023). "Emergent Reasoning Patterns in Transformer-Based Retrieval." Journal of Applied Computing, 58(2), DOI: 10.xxxx/xxxxx</i> (illustrative format only)								

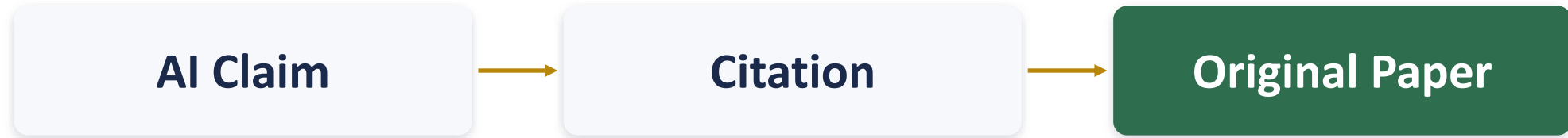
Looks scientific.

May be completely fabricated.

Professional formatting does not establish authenticity.

Citation Exists, But Does It Support the Claim?

Real paper + Real citation + Wrong interpretation = Unreliable research



Ask students to verify:

? Does the paper exist?

? Is the bibliographic information correct?

? Does it actually make / support the stated claim?

? Is the context represented correctly?

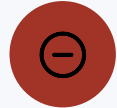
Other Risks of AI-Assisted Research



Hallucination



**Citation
fabrication**



Bias



**Outdated
information**



**Privacy /
confidentiality**



Plagiarism



Reproducibility



Overdependence

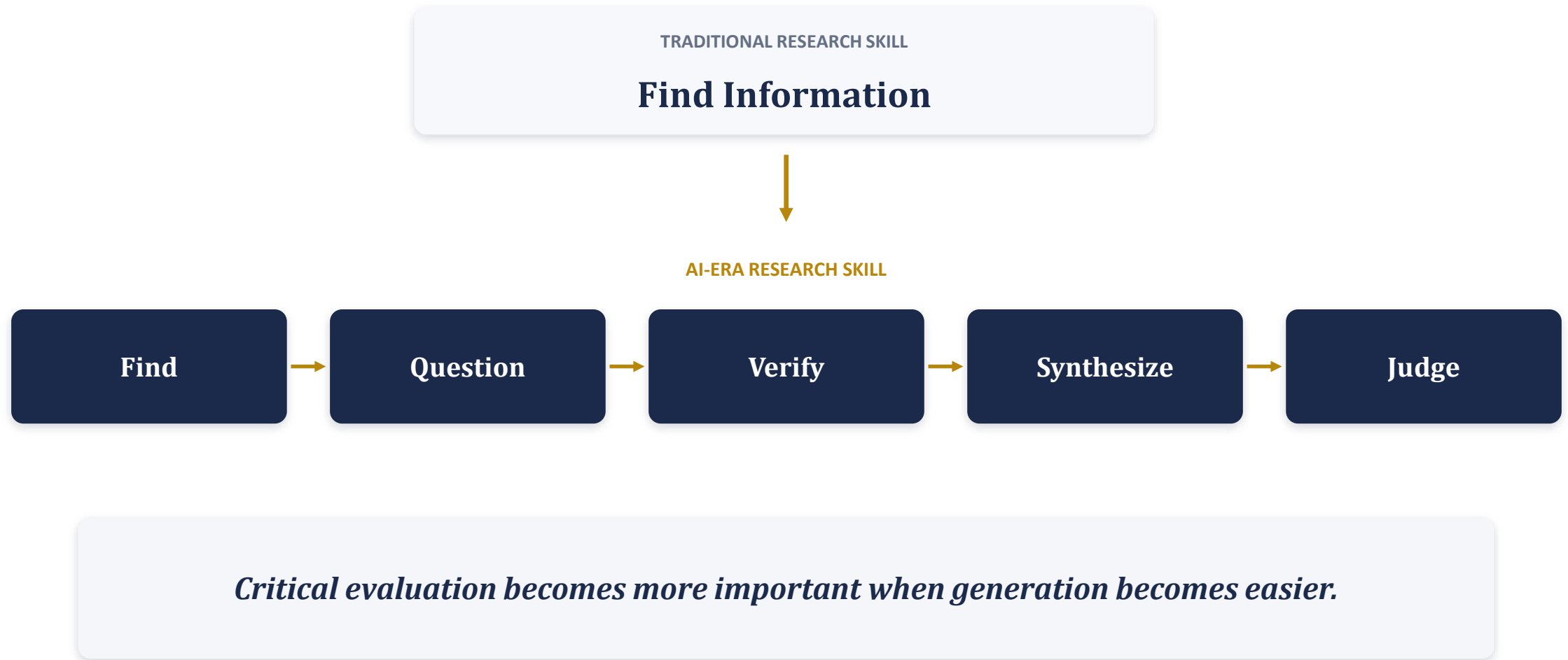


**Loss of critical
thinking**



**Unsupported
interpretation**

The Researcher's New Skill



Four Things to Remember

1

AI can assist almost every stage of research.

2

AI literacy is becoming part of research literacy.

3

AI-generated output is not scientific evidence.

4

The researcher remains responsible for verification, interpretation and integrity.

“If AI can search, summarize, analyze data, write code and draft papers, what is the role of the researcher?”

Ask better questions.

Design rigorous research.

Evaluate evidence.

Verify claims.

Interpret meaning.

Take responsibility.